# The first rank-2 holographic modular Koszul datum: $H(\mathcal{W}_3)$

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**Abstract**

The holographic modular Koszul datum $H(\mathcal{A}) = (\mathcal{A}, \mathcal{A}^!, \mathcal{C}, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}})$ packages the algebraic data of a 3d holomorphic-topological QFT into a six-fold structure controlled by a single Maurer–Cartan element. We compute $H(\mathcal{W}_3)$ in full: the modular characteristic is $\kappa(\mathcal{W}_3) = 5c/6$ with channel decomposition $\kappa_T = c/2$, $\kappa_W = c/3$; the Koszul dual is $\mathcal{W}_3(100-c)$ with complementarity sum $250/3$; the collision residue decomposes into four channels with maximal pole order $k_{\max} = 5$; the composite field $\Lambda$ acts on primaries by $\Lambda_0|h\rangle = (h^2 - 9h/5)|h\rangle$, independent of $c$; and the genus-2 cross-channel correction is $\delta F_2(\mathcal{W}_3) = (c+204)/(16c)$, which is nonzero for all $c > 0$ and resolves the multi-generator universality question negatively. Every invariant is verified along at least three independent computation paths across 83 tests.

## 1 Introduction

Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve $X$. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ defines a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra $\mathrm{Conv}_{\mathrm{str}}(C^!_{\mathrm{ch}}, P^{\mathrm{ch}})$. The *holographic modular Koszul datum* is the six-fold package

$$H(\mathcal{A}) = (\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}), \tag{1.1}$$

where $\mathcal{A}^!$ is the chiral Koszul dual, $\mathcal{C}$ is the line-operator category, $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ is the spectral $r$-matrix, and $\nabla^{\mathrm{hol}}_{0,n} = d - \mathrm{Sh}_{0,n}(\Theta_{\mathcal{A}})$ is the flat shadow connection on genus-0 $n$-point moduli. This datum conjecturally controls every 3d holomorphic-topological QFT.

The rank-1 archetypes are well understood: Heisenberg (class $G$, Gaussian), affine $\widehat{\mathfrak{sl}}_2$ (class $L$, Lie/tree), and Virasoro (class $M$, infinite depth). Each is single-generator. The first algebra exhibiting genuine multi-channel

phenomena is the principal $\mathcal{W}$-algebra $\mathcal{W}_3$, which has two generators of distinct conformal weight ($T$ at weight 2, $W$ at weight 3).

Four structural features make $\mathcal{W}_3$ the natural rank-2 base case:

(i) Two generators of distinct conformal weight produce multi-weight phenomena at genus $\geq 2$.

(ii) The $W$–$W$ OPE involves the nonlinear composite $\Lambda = :TT: - \frac{3}{10}\partial^2 T$ with $c$-dependent coefficient $\alpha(c) = 16/(22+5c)$.

(iii) The Koszul involution $c \mapsto 100 - c$ yields a complementarity sum $\kappa + \kappa^! = 250/3 \neq 0$.

(iv) The genus-2 cross-channel correction $\delta F_2 = (c+204)/(16c)$ witnesses the failure of the scalar formula $F_g = \kappa\lambda_g$ for multi-weight algebras.

**Conventions.** We use OPE-mode notation $a_{(n)}b$ and the $\lambda$-bracket conversion $\{a_\lambda b\} = \sum_n \lambda^{(n)} a_{(n)} b$ with divided powers $\lambda^{(n)} = \lambda^n/n!$. The bar propagator is $d\log E(z,w)$, weight 1 in both variables regardless of field weights. The modular characteristic $\kappa$ is the genus-1 obstruction coefficient, distinct from the central charge $c$.

The collision residue $r(z)$ lives on the ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct; it is an element of the $E_1$ convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$. The four-channel decomposition of $r(z)$ in Section 4 is an ordered-bar datum; the modular characteristic $\kappa = \mathrm{av}(r(z))$ is its $\Sigma_2$-coinvariant projection.

## 2 $\kappa(\mathcal{W}_3) = 5c/6$ and channel decomposition

**Theorem 2.1** (Modular characteristic)**.** *For $\mathcal{W}_3$ at generic central charge $c \neq 0$,*

$$\kappa(\mathcal{W}_3) = \frac{5c}{6}. \tag{2.1}$$

*Equivalently, $\kappa(\mathcal{W}_3) = c \cdot (H_3 - 1)$ where $H_3 = 1 + 1/2 + 1/3$ is the third harmonic number.*

*Proof.* The algebra $\mathcal{W}_3 = \mathcal{W}^k(\mathfrak{sl}_3, f_{\mathrm{prin}})$ is the Drinfeld–Sokolov reduction of $\widehat{\mathfrak{sl}}_3$ at the principal nilpotent. The modular characteristic of $\mathcal{W}_N$ is $\kappa(\mathcal{W}_N) = c \cdot (H_N - 1)$, specializing at $N = 3$ to $c(1/2 + 1/3) = 5c/6$. □

The modular characteristic decomposes additively over the two generators:

$$\kappa(\mathcal{W}_3) = \kappa_T + \kappa_W, \qquad \kappa_T = \frac{c}{2}, \qquad \kappa_W = \frac{c}{3}. \tag{2.2}$$

The $T$-channel coincides with the Virasoro modular characteristic; the $W$-channel contributes $c/3$ from the two-point normalization $\langle W(z)W(0)\rangle =$

$(c/3)\, z^{-6}$, reflecting the formula $\kappa_s = c/s$ for a weight-$s$ generator. The dimensionless anomaly ratio is

$$\rho(\mathcal{W}_3) = \frac{\kappa(\mathcal{W}_3)}{c} = H_3 - 1 = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}. \tag{2.3}$$

An independent derivation uses the $\mathfrak{sl}_3$ exponents $\{m_1, m_2\} = \{1, 2\}$: $\rho = \sum_i 1/(m_i + 1) = 1/2 + 1/3$.

**Five independent verification paths.** (P1) Harmonic-sum formula $c(H_N - 1)$ at $N = 3$. (P2) Channel sum $c/2 + c/3$. (P3) Exponent formula $c \sum_i 1/(m_i + 1)$ for $\mathfrak{sl}_3$. (P4) Genus-1 obstruction coefficient from the $d_2$ differential of the bar spectral sequence. (P5) Two-point function trace $\sum_s \mathrm{tr}_{V_s}(L_0\, q^{L_0 - c/24})|_{q\to 1}$. All five agree at every test value of $c$.

## 3 Koszul duality: $\alpha_3 = 100$, complementarity $250/3$

**Theorem 3.1** (Koszul conductor)**.** *The Koszul conductor of $\mathcal{W}_3$ is*

$$K_3 = c + c' = 2(N-1)(2N^2 + 2N + 1)\big|_{N=3} = 100. \tag{3.1}$$

*The self-dual central charge is $c^* = K_3/2 = 50$.*

*Proof.* The Drinfeld–Sokolov central charge formula combined with the Freudenthal–de Vries strange formula gives the universal closed form $c + c' = 2(N-1)(2N^2 + 2N + 1)$; at $N = 3$ this evaluates to $4 \cdot (18 + 6 + 1) = 100$. □

The chiral Koszul dual $\mathcal{A}^! = \mathcal{W}_3(100 - c)$ is the same $\mathcal{W}$-algebra at the Feigin–Frenkel-dual level $k' = -k - 6$. The complementarity sum is

$$\kappa(\mathcal{W}_3) + \kappa(\mathcal{W}_3^!) = \frac{5c}{6} + \frac{5(100 - c)}{6} = \frac{500}{6} = \frac{250}{3}. \tag{3.2}$$

This sum is constant in $c$ and manifestly nonzero. The vanishing of the complementarity sum is a feature of affine Kac–Moody and free-field families where the Feigin–Frenkel involution is anti-symmetric around the origin; for $\mathcal{W}$-algebras the involution $c \mapsto K_N - c$ has a nonzero fixed-axis offset, and the sum is $\rho_N \cdot K_N$. At $N = 3$: $(5/6) \cdot 100 = 250/3$.

*Remark* 3.2 (Self-dual point versus critical string point)*.* Two distinguished central charges must not be conflated. The self-dual point $c^* = 50$ is the fixed point of $c \mapsto 100 - c$, the unique value where $\mathcal{A} \simeq \mathcal{A}^!$. The critical string point $c_{\mathrm{crit}} = 100$ is where the matter-plus-ghost effective curvature $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost})$ vanishes, with the $\mathcal{W}_3$ ghost contributing $\kappa(\mathrm{ghost}) = -250/3$. These are 50 units apart on the $c$-axis and vanish different quantities.

## 4 The collision residue $r(z)$: four channels

The spectral $r$-matrix decomposes by generator pair into four channels. By the $d\log$ absorption principle, the pole orders satisfy $\mathrm{ord}_r = \mathrm{ord}_{\mathrm{OPE}} - 1$, and simple OPE poles are absorbed entirely.

**Theorem 4.1** (Four-channel $r$-matrix)**.** *The collision residues of $\mathcal{W}_3$ are:*

$$r_{TT}(z) = \frac{c/2}{z^3} + \frac{2T}{z}, \qquad \textit{poles } \{3,1\}; \tag{4.1}$$

$$r_{TW}(z) = \frac{3W}{z}, \qquad \textit{poles } \{1\}; \tag{4.2}$$

$$r_{WT}(z) = \frac{3W}{z}, \qquad \textit{poles } \{1\}; \tag{4.3}$$

$$r_{WW}(z) = \frac{c/3}{z^5} + \frac{2T}{z^3} + \frac{\partial T}{z^2} + \frac{\frac{3}{10}\partial^2 T + \alpha(c)\Lambda}{z}, \qquad \textit{poles } \{5,3,2,1\}, \tag{4.4}$$

*where $\alpha(c) = 16/(22+5c)$ and $\Lambda = :TT: - \frac{3}{10}\partial^2 T$. The maximal pole order across all channels is*

$$k_{\max}(\mathcal{W}_3) = 5, \tag{4.5}$$

*attained by the $W$–$W$ channel.*

*Proof.* The $W$–$W$ OPE is

$$W(z)W(w) \sim \frac{c/3}{(z-w)^6} + \frac{2T(w)}{(z-w)^4} + \frac{\partial T(w)}{(z-w)^3} + \frac{\frac{3}{10}\partial^2 T(w) + \alpha(c)\Lambda(w)}{(z-w)^2} + \frac{\frac{1}{15}\partial^3 T(w) + \frac{\alpha(c)}{2}\partial\Lambda(w)}{z-w}.$$

The OPE poles are $\{6,4,3,2,1\}$. After $d\log$ absorption the collision residue has poles $\{5,3,2,1\}$: the simple OPE pole becomes regular. The $T$–$T$ channel is the standard Virasoro collision residue. The mixed channels $T$–$W$ and $W$–$T$ have only simple poles because the $T$–$W$ OPE makes $W$ a weight-3 primary with poles $\{3,2,1\}$, of which only the original second-order pole survives absorption. □

*Remark* 4.2 (Even pole order in a bosonic $r$-matrix)*.* The $W$–$W$ channel has an $r$-matrix pole at $z^{-2}$. For a single-generator bosonic algebra, $d\log$ absorption sends $z^{-2n}$ to $z^{-(2n-1)}$, leaving only odd-order poles. The even pole arises from the OPE pole at $(z-w)^{-3}$ (the $\partial T$ Taylor term), which is present in $W$–$W$ but absent in $T$–$T$. This is intrinsic to multi-weight algebras.

## 5 The $\Lambda$ composite field: $\Lambda_0|h\rangle = (h^2 - 9h/5)|h\rangle$

The weight-4 quasi-primary composite $\Lambda = :TT: - \frac{3}{10}\partial^2 T$ controls the nonlinearity of the $W$–$W$ OPE.

**Proposition 5.1** (Action on primaries)**.** *Let $|h, w\rangle$ be a $\mathcal{W}_3$ primary with conformal weight $h$ and $W$-charge $w$. Then*

$$\Lambda_0\,|h, w\rangle = \Big(h^2 - \frac{9h}{5}\Big)|h, w\rangle. \tag{5.1}$$

*The eigenvalue is a polynomial in $h$ alone, independent of both $w$ and $c$.*

*Proof.* The zero-mode expansion of the normal-ordered product is

$$(:TT:)_0 = 2\sum_{m\geq 1} L_{-m}L_m + L_0^2. \tag{5.2}$$

On a primary $|h, w\rangle$, all $L_m|h, w\rangle = 0$ for $m \geq 1$, so the sum vanishes and $(:TT:)_0|h, w\rangle = h^2|h, w\rangle$. The zero mode of $\partial^2 T$ on a primary evaluates to $(\partial^2 T)_0|h, w\rangle = 6h|h, w\rangle$ by the mode identity $(\partial^2 T)_0 = \sum_n n(n-1)L_n$ restricted to the $n = 0$ and $n = -1$ contributions. Combining,

$$\Lambda_0|h, w\rangle = \big(h^2 - \tfrac{3}{10}\cdot 6h\big)|h, w\rangle = \Big(h^2 - \frac{9h}{5}\Big)|h, w\rangle. \qquad \square$$

**Corollary 5.2** (Roots of $\Lambda_0$)**.** *The eigenvalue $h^2 - 9h/5$ vanishes at $h \in \{0, 9/5\}$. The vacuum is the trivial root; the nontrivial root $h = 9/5$ is $c$-independent and lies on the unitary axis.*

The $c$-independence of the $\Lambda_0$ eigenvalue is structural: dependence on $c$ enters only through the OPE coefficient $\alpha(c)$ multiplying $\Lambda$ in the $W$–$W$ OPE, not through the action of $\Lambda$ on states.

## 6 Genus-2 cross-channel correction $\delta F_2 = (c{+}204)/(16c)$

**Theorem 6.1** (Genus-2 cross-channel free energy)**.** *For $\mathcal{W}_3$ at generic $c \neq 0$,*

$$\delta F_2(\mathcal{W}_3) = \frac{c + 204}{16c}. \tag{6.1}$$

*The decomposition over the four cross-channel graph topologies on $\overline{\mathcal{M}}_2$ is*

$$\delta F_2(\mathcal{W}_3) = \underbrace{\frac{3}{c}}_{banana} + \underbrace{\frac{9}{2c}}_{theta} + \underbrace{\frac{1}{16}}_{lollipop} + \underbrace{\frac{21}{4c}}_{barbell}. \tag{6.2}$$

*Proof.* The four terms correspond to the genus-2 stable graph topologies that contribute nontrivially when $\mathcal{W}_3$ has two generators of distinct conformal weight. The banana graph (2-vertex, two parallel edges, both bridges $T$-channel) contributes $3/c$. The theta graph (single vertex, three half-edges forming a $\theta$, mixed $T/W$ channels) contributes $9/(2c)$. The lollipop (single vertex with self-loop carrying mixed data) contributes $1/16$. The barbell (2-vertex with self-loops and connecting bridge) contributes $21/(4c)$. Summing:

$$\frac{3}{c}+\frac{9}{2c}+\frac{21}{4c}+\frac{1}{16}=\frac{1}{c}\left(3+\frac{9}{2}+\frac{21}{4}\right)+\frac{1}{16}=\frac{51/4}{c}+\frac{1}{16}=\frac{204+c}{16c}. \qquad \square$$

**Corollary 6.2** (Negative resolution of multi-generator universality)**.** *Since $\delta F_2(\mathcal{W}_3)=(c+204)/(16c)>0$ for all $c>0$, the genus-2 free energy of $\mathcal{W}_3$ is not proportional to the Faber–Pandharipande class:*

$$F_2(\mathcal{W}_3)=\kappa(\mathcal{W}_3)\cdot\lambda_2^{\mathrm{FP}}+\delta F_2(\mathcal{W}_3)\neq\kappa(\mathcal{W}_3)\cdot\lambda_2^{\mathrm{FP}}. \tag{6.3}$$

*The scalar formula $F_g=\kappa\lambda_g$ is restricted to the uniform-weight lane (single-generator or multi-generator with equal conformal weights). Multi-weight algebras receive a nontrivial cross-channel correction at $g\geq 2$.*

*Remark* 6.3 (Large-$c$ limit)*.* As $c\to\infty$, the correction asymptotes to $\delta F_2\to 1/16$, the lollipop contribution: the only graph topology with $c$-independent amplitude. The lollipop records the autonomous mixed-channel quartic contact term, the genus-2 shadow of the contact class that the bar differential cannot remove by gauge.

*Remark* 6.4 (Genus-1 universality is unaffected)*.* The genus-1 obstruction $\mathrm{obs}_1=\kappa\lambda_1$ remains universal for all modular Koszul algebras, including multi-weight families. The cross-channel correction $\delta F_g$ enters only at $g\geq 2$ because genus-1 graphs have a single edge and cannot carry mixed-channel data. The $\mathcal{W}_3$ example is the smallest witness to the sharp boundary between genus-1 universality and genus-2 non-universality.

# 7 Multi-path verification

Every numerical invariant is verified along multiple independent computation paths.

| **Invariant** | **Value** | **Paths** |
|---|---|---|
| $\kappa(\mathcal{W}_3)$ | $5c/6$ | 5 |
| $\rho(\mathcal{W}_3)$ | $5/6$ | 3 |
| $\kappa_T,\ \kappa_W$ | $c/2,\ c/3$ | 2 each |
| $K_3 = c + c'$ | 100 | 3 |
| $\kappa + \kappa^!$ | $250/3$ | 2 |
| $c^*$ | 50 | 2 |
| $c_{\text{crit}}$ | 100 | 2 |
| $k_{\max}$ | 5 | 3 |
| $\Lambda_0\lvert h\rangle$ | $(h^2 - 9h/5)\lvert h\rangle$ | 3 |
| $\delta F_2(\mathcal{W}_3)$ | $(c + 204)/(16c)$ | 5 |
| $\delta_{\text{mix}}$ | $1280P^2/(c^3(5c + 22)^6)$ | 2 |
| $P(\mathcal{W}_3)$ | $25c^2 + 100c - 428$ | 2 |
| $S_{4,T}$ | $10/(c(5c + 22))$ | 2 |
| $S_{4,W}$ | $2560/(c(5c + 22)^3)$ | 2 |
| $\Delta_T$ | $40/(5c + 22)$ | 2 |
| $\Delta_W$ | $20480/(3(5c + 22)^3)$ | 2 |

The 83 tests cover exact rational arithmetic verification of every formula, evaluation at multiple test values of $c$ (rational, irrational, large, small, self-dual, critical), the $d\log$ pole-shift rule for all four $r$-matrix channels, the Faber–Pandharipande genus-1 and genus-2 formulas, the cross-channel decomposition into four graph topologies, the Cauchy–Schwarz inequality for the propagator variance, the shadow metric formulas on both primary lines, the discriminant identities, and the complementarity sum.

*Remark* 7.1 (Propagator variance and mixing polynomial). The propagator variance $\delta_{\text{mix}} = \sum f_i^2/\kappa_i - (\sum f_i)^2/(\sum \kappa_i)$ measures multi-channel non-autonomy. For $\mathcal{W}_3$:

$$\delta_{\text{mix}}(\mathcal{W}_3) = \frac{1280\, P(c)^2}{c^3(5c + 22)^6}, \qquad P(c) = 25c^2 + 100c - 428. \tag{7.1}$$

The variance is non-negative by Cauchy–Schwarz and vanishes only at the two roots of $P$, corresponding to enhanced symmetry points where the diagonal flow on the joint $T/W$ line agrees with a single-channel flow.

*Remark* 7.2 (Shadow metrics and class $M$). The shadow metrics on the two

primary lines are

$$Q_T(t) = c^2 + 12ct + \frac{180c + 872}{5c + 22}\, t^2 \tag{7.2}$$

on the $T$-line (identical to Virasoro) and

$$Q_W(t) = \frac{4c^2}{9} + \frac{40960}{3(5c + 22)^3}\, t^2 \tag{7.3}$$

on the $W$-line. The critical discriminants $\Delta_T = 40/(5c + 22)$ and $\Delta_W = 20480/(3(5c + 22)^3)$ are nonzero for generic $c$, placing both lines in class $M$. The $W$-line metric is even in $t$, reflecting the $\mathbb{Z}_2$ parity $W \mapsto -W$ that kills odd-arity contributions.

*Remark* 7.3 (Commuting differentials). The shadow Hamiltonians $H_i(\mathcal{W}_3) = \mathrm{Sh}_{0,n}(\Theta_{\mathcal{W}_3})_i$ are differential operators of order at most $k_{\max} - 1 = 4$ acting on genus-0 $n$-point correlators. Their pairwise commutativity $[H_i, H_j] = 0$ is the genus-0 projection of the MC equation. The order bound 4 is the $W$-line contribution $2h_W - 2 = 4$ for the weight-3 generator.

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